

CED 101: Introduction to Chemical Engineering

Course Code: CED 101

Course Name: Introduction to Chemical Engineering

L-T-P-C: 2-1-0-3

Prerequisites: None

Intended for: B.Tech. Chemical Engineering with Data Analytics

Mutual Exclusion: None

Approval: 65th BoA

Preamble

This foundational course introduces first-year students to the fundamentals and scope of Chemical Engineering within the interdisciplinary framework of the Chemical Engineering with Data Analytics program. Students gain an overview of chemical process industries, chemical plants, unit operations, and unit processes, along with an introduction to the core areas of chemical engineering. The course also presents the role of Data Analytics in modern chemical engineering, highlighting how data-driven methods and computational tools enable process optimization, predictive decision-making, and accelerated materials and process discovery. It establishes the foundation for integrating chemical engineering principles with data-centric approaches throughout the curriculum.

Course Modules

- **Module I: Introduction to Chemical Engineering:** Scope of chemical engineering, Introduction to various chemical industries, Role of chemical engineers in industry and society (Lectures: 1)
- **Module II: Unit operations and unit processes in the chemical industry:** Introduction to various unit operations in downstream chemical processing, with some examples in inorganic and organic chemical manufacture (Lectures: 3)
- **Module III: Chemical Process Calculations:** Introduction to units and dimensions of quantities, basic materials balances (single phase, multiphase systems), Energy balances (nonreactive and reactive systems, Fuels and combustion), Hess's law, heats of formation and reaction (Lectures: 4, Tutorials: 2)
- **Module IV: Transport processes: Momentum transfer (Fluid mechanics):** Nature of fluids, Newton's law of viscosity, Equation of motion, Equation of continuity, Newton's law of viscosity, Fluid flow past solid surface, Flow through tubes, Frictional losses, pressure drop, Flow through porous media (Lectures: 3, Tutorials: 2)
- **Module V: Transport processes: Heat transfer:** Basic concepts of conduction, convection, radiation, introduction to heat exchangers (double pipe, shell-and-tube), evaporation (Lectures: 3, Tutorials: 2)
- **Module VI: Transport processes: Mass Transfer:** Basics of diffusion (Fick's law), interfacial mass transfer, vapor-liquid equilibrium, relative volatility, basic concepts of distillation and absorption, liquid-liquid extraction, equipment for gas-liquid operations, Basic concepts of simultaneous momentum, heat and mass transfer (transport phenomena) (Lectures: 3, Tutorials: 2)
- **Module VII: Chemical reaction engineering (chemical reactors and kinetics):** Introduction to batch and continuous flow (continuous-stirred tank, tubular, packed-bed) reactors, reactor sizing (design equa-

tions), rate laws and stoichiometry, collection and analysis of rate data, multiple reactions (parallel and series) (Lectures: 4, Tutorials: 2)

- **Module VIII: Chemical Engineering Thermodynamics:** Thermodynamic systems, Variables and processes, Equilibrium and phase rule, Equations of state, first law of thermodynamics and its applications, second law of thermodynamics, internal energy and enthalpy, second law of thermodynamics, power and refrigeration cycles (Lectures: 3, Tutorials: 2)
- **Module IX: Introduction to Chemical Engineering Computing:** Scientific notations, Significant figures and precision, Measures of central tendency and dispersion (mean, mode, median, range, standard deviation, skewness), Data visualization (charts and plots, logarithmic coordinates), Process data analysis and representation, Analysis of variance, Curve fitting (linear and nonlinear), Hypothesis testing and confidence intervals (Lectures: 4, Tutorials: 2)

Text Books

1. J. M. Coulson, J.F. Richardson, J. H. Harker and J. R. Backhurst, *Coulson & Richardson's Chemical Engineering*, Vol. 1, 6th Edition, Butterworth-Heinemann, 1999.
2. R. M. Felder, R. W. Rousseau, *Elementary principles of chemical processes*, 4th Ed., John Wiley and Sons, 2017.

References

1. S.K. Ghosal, S.K. Sanyal and S. Datta, *Introduction to Chemical Engineering*, McGraw Hill, 1993.
2. Y. V. C. Rao, *Chemical Engineering Thermodynamics*, Universities Press, 1997.
3. V. M. Zavala, *Statistics for Chemical Engineers: From Data to Models to Decisions*, Cambridge University Press, 2025.
4. Holman, J. P. (2010). *Heat transfer* (10th ed.). McGraw-Hill.
5. Fogler, H. S. (2016). *Elements of chemical reaction engineering* (5th ed.). Prentice Hall.
6. Rao, M. G., & Sittig, M. (Eds.). (2018). *Dryden's outlines of chemical technology for the 21st century* (3rd ed.). Semantic Scholar
7. S. Pushpavanam, *Introduction to Chemical Engineering*, Prentice Hall of India, New Delhi, 2012
8. Spiegel, M.R., Schiller, J.J. and Srinivasan, R.A. (2012) *Schaum's Outline of Probability and Statistics*, 4th edn. New York: McGraw-Hill.